

AWS Certified Generative AI Developer — Professional

The Use Case

- Healthcare chatbot built on Bedrock foundation models, giving medical information to patients
- Bias dimensions to evaluate:** age (elderly vs younger), gender and gender identity, ethnic and cultural background, socioeconomic factors, **medical literacy levels**

The last dimension is the least obvious: if the chatbot answers in clinical language, it is useless to a patient with low medical literacy. **That is a fairness failure too** — the same answer is not equally useful to everyone.

How Domain 3's four tasks differ

Task	Question
3.1	Is the content safe?
3.2	Is the data safe?
3.3	Can you prove it? (process, audit, lineage)
3.4	Is it right for everyone? (transparent, fair, ethical)

3.3 vs 3.4 is the pair worth separating: governance proves you followed the *process*; responsible AI concerns whether the *outcome* is actually good.

1. Transparent AI Systems

Bedrock agent tracing

```
InvokeAgent API with the tracing parameter enabled
-> agent execution steps and reasoning paths
-> knowledge base retrievals + RELEVANCE SCORES
-> API calls made during execution
-> decision points and confidence scores at each step
```

In Task 2.1 the agent's ReAct loop was invisible. **Tracing opens that box** — useful both for user-facing transparency and for debugging why an agent went wrong.

Reasoning displays

```
CoT prompting produces a reasoning chain
-> custom UI components render it
-> the user sees how the conclusion was reached
```

Task 1.6 advised that reasoning need not be shown to the user. **Here it is deliberately shown** — the goal has changed from accuracy to transparency. With **streaming**, intermediate reasoning appears in real time.

Confidence metrics

Technique	How
Token-level probability analysis	Low token probabilities indicate uncertainty
Ensemble methods	Compare several models to derive confidence intervals
Calibration	Does "90% confident" actually mean right 90% of the time? Models are typically over-confident, so calibration aligns stated confidence with real accuracy.

Tracked over time as CloudWatch custom metrics.

Source attribution — advanced

```
identify automatically when output derives from a source
RAG with EXPLICIT citation mechanisms
track attribution THROUGH multiple reasoning steps
provide verifiable links to original materials
```

The third point is the hard one: if the model reasoned across three steps, the citation from step 1 must still be attached to the conclusion at step 3. In healthcare this matters — the patient should know which clinical guideline the advice came from.

Explainability visualisation

step-by-step reasoning breakdowns with confidence metrics
citation networks linking content to sources
uncertainty visualisation (confidence intervals)
alternative reasoning path exploration

Transparency metrics

Metric	Indicates
Citation frequency	How often sources are cited
Reasoning step count	How much reasoning is exposed
Uncertainty acknowledgment	How often the model says it doesn't know

Admitting uncertainty is a **signal of transparency**. A model that never expresses doubt is over-confident, not knowledgeable.

2. Fairness Evaluations

Fairness metrics — they measure different things

Metric	Definition
Demographic parity	Every group receives positive outcomes at the same rate. Simple, but forcing equality can itself be wrong when groups genuinely differ.
Equal opportunity	Among those who qualify , all groups receive positive outcomes at the same rate
Equalized odds	Both true positive rate <i>and</i> false positive rate match across groups — the strictest
Counterfactual fairness	Change only the demographic attribute; the answer must not change
Group vs individual	Group: average outcomes across groups. Individual: similar people receive similar treatment.

```
"a 65-year-old woman with chest pain"  -> answer A
"a 65-year-old man with chest pain"      -> answer B

A differs from B  ->  bias
```

Counterfactual testing is the most practical test for a GenAI system, because it isolates the causal effect of the demographic attribute.

LLM-as-a-Judge

```
build a diverse test dataset across demographics
-> collect chatbot responses
-> a judge model scores them
-> for bias, stereotyping, unfair treatment
```

Transcript's metrics: **correctness, completeness, harmfulness.**

Completeness is the subtle one: the model may answer every group correctly but give some groups **less detail**. That is also bias.

The judge has its own problems:

```
carefully designed RUBRICS          -> consistency
ensemble judging                     -> multiple judge models
control for the JUDGE'S OWN BIAS
calibrate against HUMAN evaluations
```

The judge is itself an LLM. If it treats plain language as lower quality, it will penalise exactly the medical-literacy fairness you are trying to achieve — which is why human calibration is required.

Systematic bias testing

```
test across diverse demographic dimensions (not one axis)
counterfactual testing to isolate causal bias
statistical significance testing  -> is the gap real or noise?
track bias metrics over time      -> drift or improvement
```

A/B testing with Prompt Management and Prompt Flows

```
Prompt Management -> multiple prompt variants, each designed
                    to reduce bias for a specific group
Prompt Flows       -> route questions dynamically
                    -> test strategies across demographic segments
                    -> evaluate against fairness metrics
Visual builder     -> link models, prompts and AWS services
                    -> select the fairest effective strategy
```

Advanced consideration	Why
Control confounding variables	Vary one prompt element at a time, or you cannot attribute the effect
Stratified sampling	Every demographic group properly represented in the test set
Statistical power analysis	Determine how large the test must be to trust the result
Interaction effects	Model parameters can affect groups differently — e.g. temperature changes may hit some groups harder

Test datasets and human evaluation

- Fairness datasets must represent **diverse perspectives and edge cases**. This is the most labour-intensive part of the work and the most consequential — you can only detect the bias your test set can express.
- Human evaluation supplements automated testing** — own workforce or AWS-managed. The LLM judge says it is fair; only a

human can confirm a real patient would agree. Automation gives scale, humans give ground truth.

Continuous fairness monitoring

```
custom fairness metrics per demographic dimension
-> dashboards over time and across model versions
-> baselines established
-> alarms when thresholds are exceeded
-> notifications or remediation workflows
```

```
also monitor: invocations and TOKEN COUNTS per user segment
```

Token count is a proxy for completeness bias — if one demographic segment consistently receives shorter answers, that is a fairness signal that no accuracy metric would surface.

Plus **automated fairness auditing** on a schedule, producing compliance reports, and a **feedback loop** that feeds evaluation results back into prompt and model improvement.

3. Policy-Compliant AI Systems

Policy translation — the core skill

create formal specifications of policy requirements

-> MAP requirements to specific technical implementations

-> build verification systems that can PROVE compliance

-> maintain TRACEABILITY between policy and implementation

POLICY: "The AI must not discriminate against any group"

| translate

TECHNICAL:

counterfactual fairness test on every release

DI metric held within 0.8-1.25

CloudWatch alarm when breached

Guardrails filters for biased language

Traceability matters because in an audit you must show which policy each control implements — otherwise you cannot demonstrate the policy was ever put into effect.

Advanced guardrail configuration

Technique	Purpose
Context-aware guardrails	Adapt to different use cases
Multi-stage filtering pipelines	Different sensitivity levels at different stages
Balancing safety with utility	The hardest trade-off — too strict and a medical chatbot cannot discuss medicine
Regulatory-specific guardrails	Implement particular regulatory requirements

content filters

topic filters

custom blocklists

CONTEXTUAL GROUNDING

-> detect and block biased language

-> factual accuracy checks

Transcript adds a two-layer approach: use the LLM provider's **built-in guardrails** *and* deploy **external LLM-based guardrails** for additional protection.

Model Cards — the responsible AI angle

document FM limitations

document known BIASES

intended use cases

performance across scenarios AND demographic groups

Documenting known biases rather than hiding them is itself a responsible AI practice. And performance must be reported **per demographic group** — an overall 92% can conceal 70% for one group.

Cards update automatically as new evaluation data arrives (Task 3.3's CI/CD integration).

Automated compliance checking — Lambda

rule engines evaluating content against policy

-> real-time analysis of inputs and outputs

-> different rules depending on CONTEXT and use case

-> detailed compliance reports WITH EVIDENCE

-> integration with remediation workflows

"With evidence" matters — a pass/fail verdict is not enough; the reason must be recorded.

Multi-layered policy enforcement

pre-processing

in-processing

post-processing

-> guardrails

-> monitoring

-> filters

Task 3.1's defense-in-depth, applied to policy rather than safety.

Continuous compliance monitoring

CloudWatch metrics and alarms track adherence over time, with dashboards showing compliance status and alerts on violations **or drift from established standards** — the gradual kind of failure that happens over months rather than in a single incident.

AWS Audit Manager

```
AWS Audit Manager
+ GenAI Best Practices Framework
-> document fairness evaluations
-> maintain compliance evidence
```

What it does: automates **evidence collection** against controls, so you are not manually gathering screenshots and logs before an audit. The **GenAI Best Practices Framework** is a pre-built control set specific to generative AI — worth remembering, since it is purpose-built for this exam's subject matter.

Version control for policies

Guardrail and policy configurations are versioned, reviewed and deployed through governance processes — the same discipline applied to prompts in Task 1.6, because **changing a guardrail is itself a governance decision**.

The full picture

```
Organizational Policy
| translate + traceability
Technical Controls
+-- Guardrails (context-aware, multi-stage)
+-- Lambda compliance checks (rule engines)
+-- Model Cards (limitations, biases, per-group performance)
+-- Fairness tests (counterfactual, LLM-as-judge)
|
CloudWatch continuous monitoring
|
Audit Manager -> automated evidence collection
|
Compliance reports
```

Service limits and feature availability change. Verify specific numbers against current AWS documentation before the exam.